Pulseq interpreter and workflow on GE

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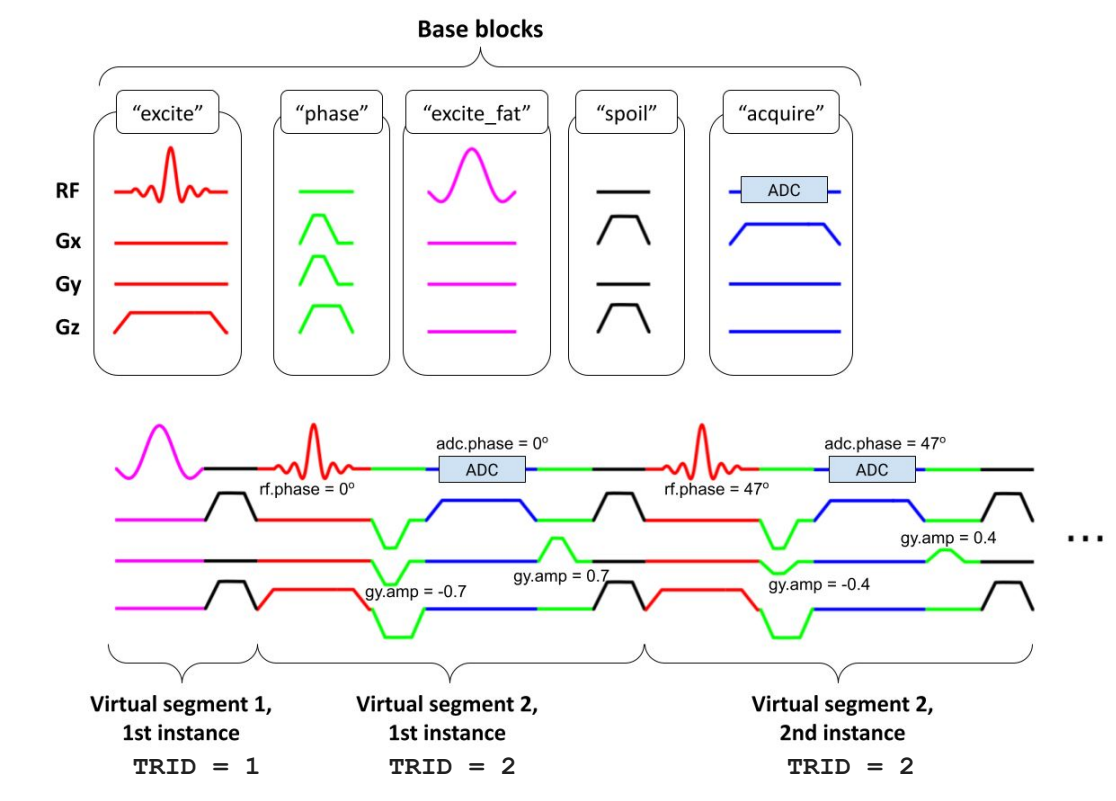
Tues Feb 24, 2026

ISMRM Virtual Meeting: Vendor-Agnostic Pulse Sequence Programming with Pulseq II

GE: Segmented (“TR”) sequence representation



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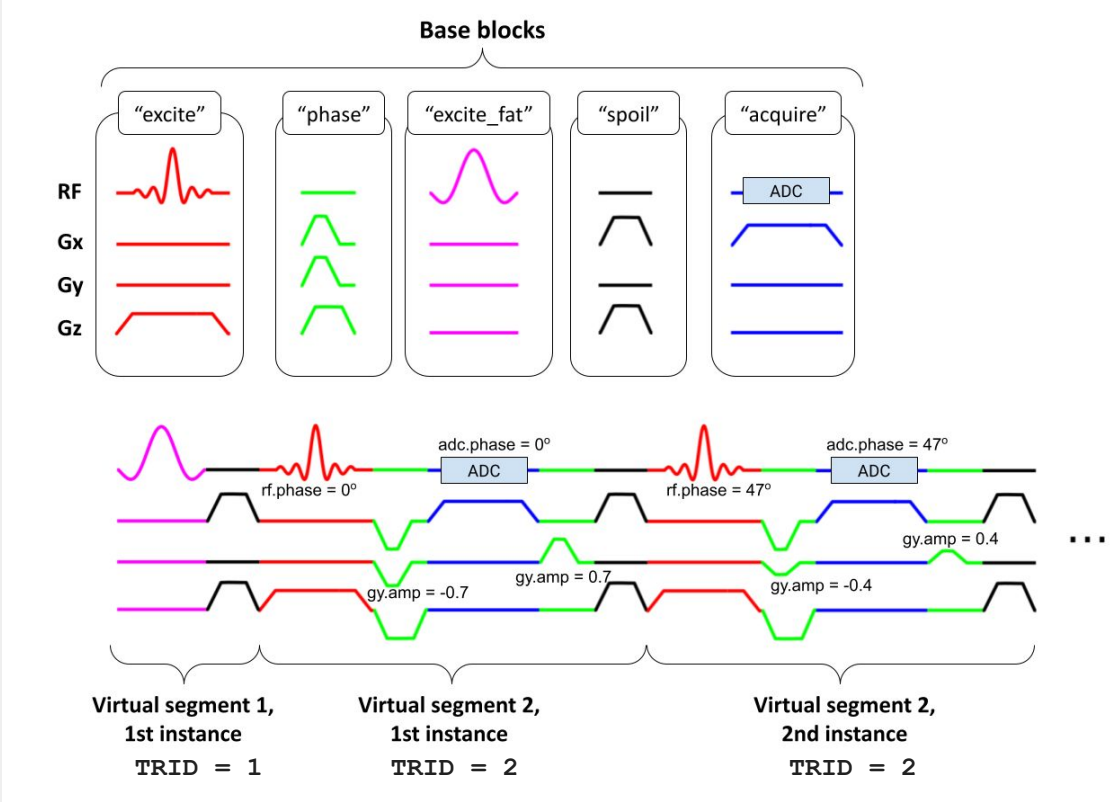


NEW

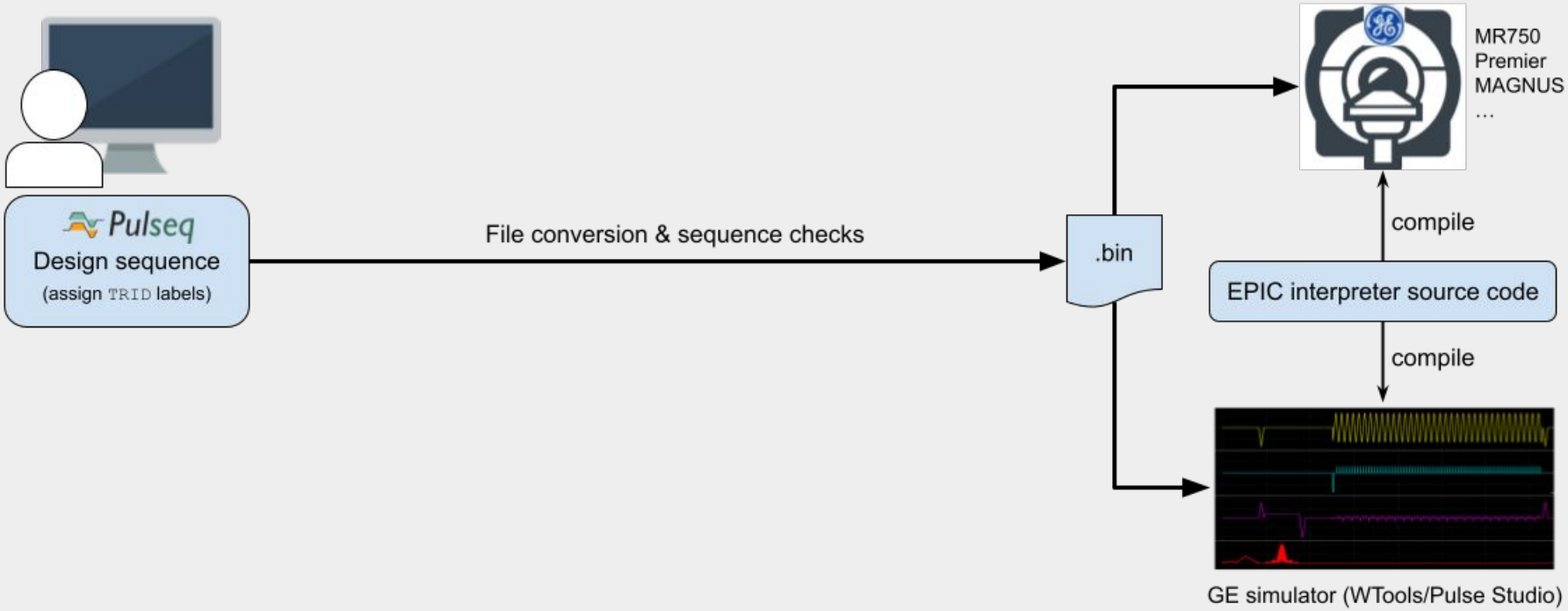
PulSeg

https://github.com/HarmonizedMRI/pulseg/

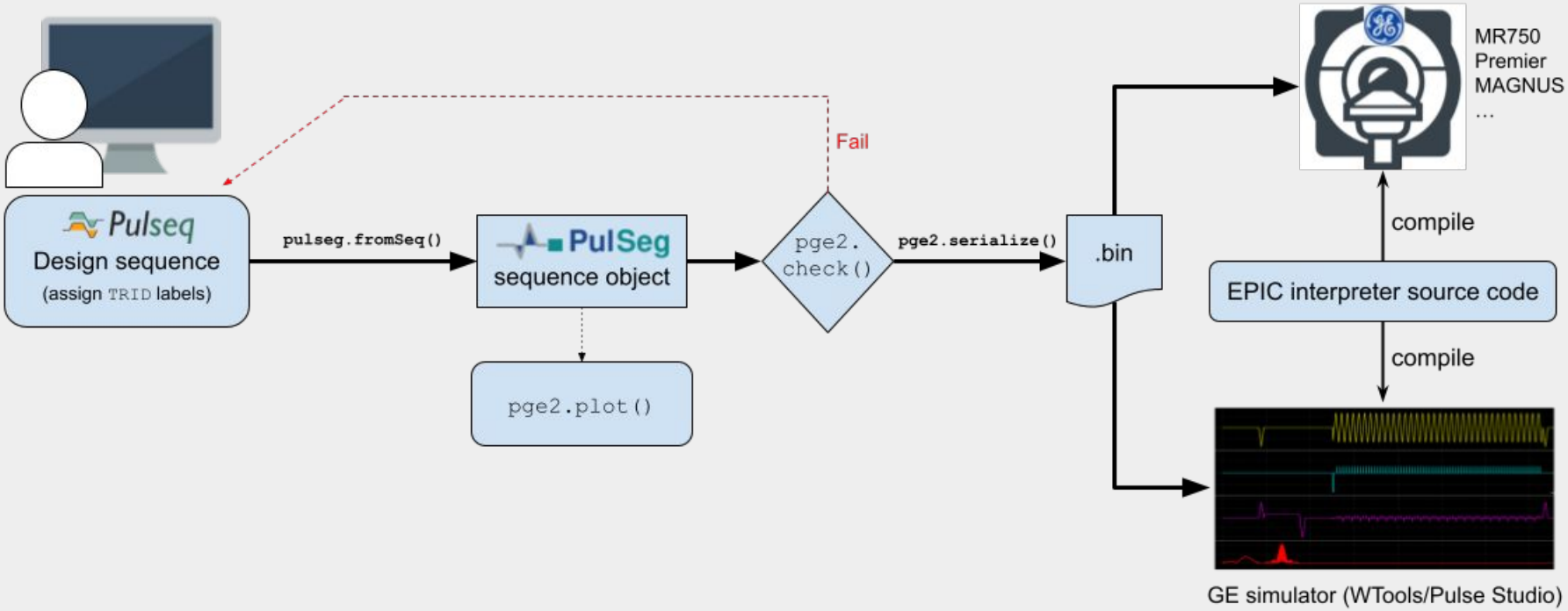
```
struct PulsegSequence
    baseBlocks: [1x8 struct]
    virtualSegments: [1x4 struct]
    scanLoop: [2462x23 double]
end

psq = pulseg.fromSeq('gre.seq');
```

(Formerly “seq2ceq.m” in PulCeq toolbox)

Workflow


Pulseq
Design sequence
(assign TRID labels)
File conversion & sequence checks
.bin
MR750
Premier
MAGNUS
...
compile
EPIC interpreter source code
compile
GE simulator (WTools/Pulse Studio)

Workflow


Pulseq
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(assign TRID labels)
pulseg.fromSeq()
PulSeg
sequence object
pge2.plot()
Fail
pge2.check()
pge2.serialize()
.bin
MR750
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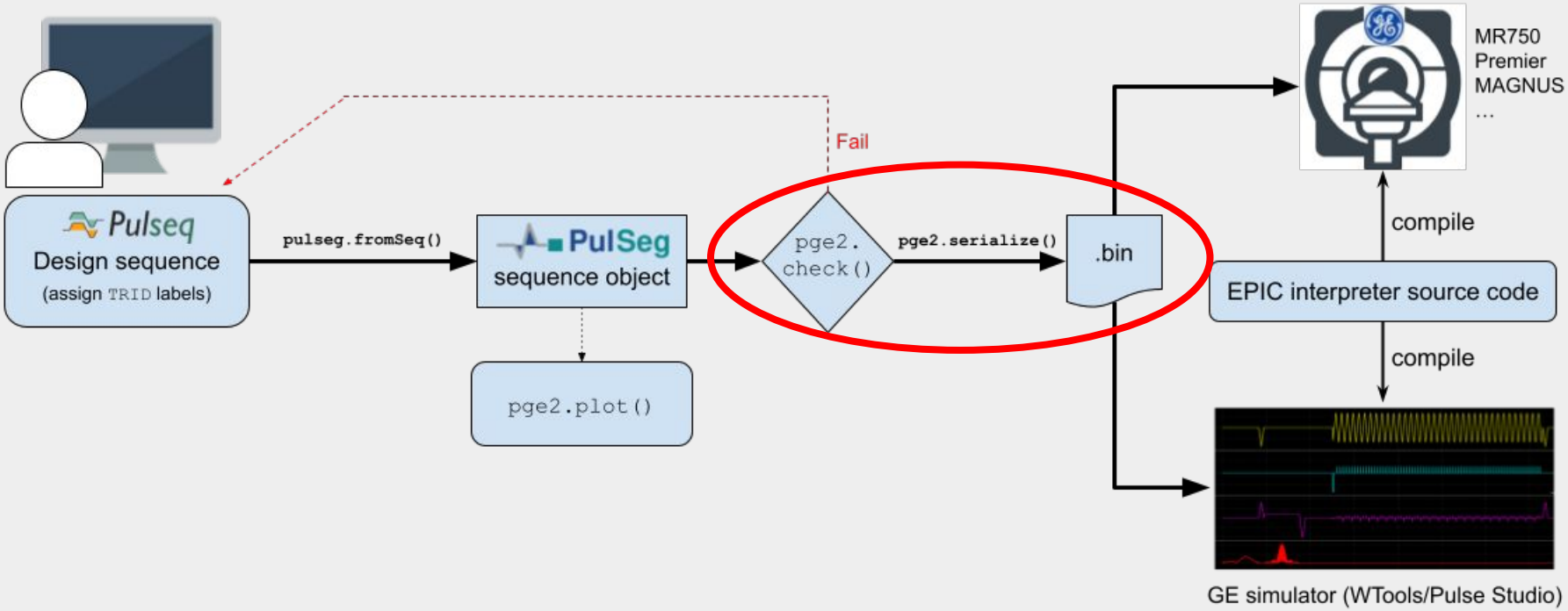
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Serialize PulSeg object for execution on GE scanners

```
%% Load sequence
psq = pulseg.fromSeq('gre.seq');

%% Define scanner hardware (MR750 example)
psd_rf_wait  = 100e-6;   % RF-gradient delay (s)
psd_grd_wait = 80e-6;    % ADC-gradient delay (s)
b1_max   = 0.25;         % Gauss
g_max    = 5;            % Gauss/cm
slew_max = 20;           % Gauss/cm/ms
coil     = 'xrm';        % Coil name for PNS checks

sysGE = pge2.opts(psd_rf_wait, psd_grd_wait, b1_max, g_max, slew_max, coil);

%% Run PNS and hardware compatibility checks
PNSwt = [0.8 1 0.7];   % directional PNS weights, see pge2.pns()
params = pge2.check(psq, sysGE, 'PNSwt', PNSwt);

%% Export to binary file
pge2.serialize(psq, 'output.bin', 'sysGE', sysGE, 'params', params, 'pislquant', 64);
```

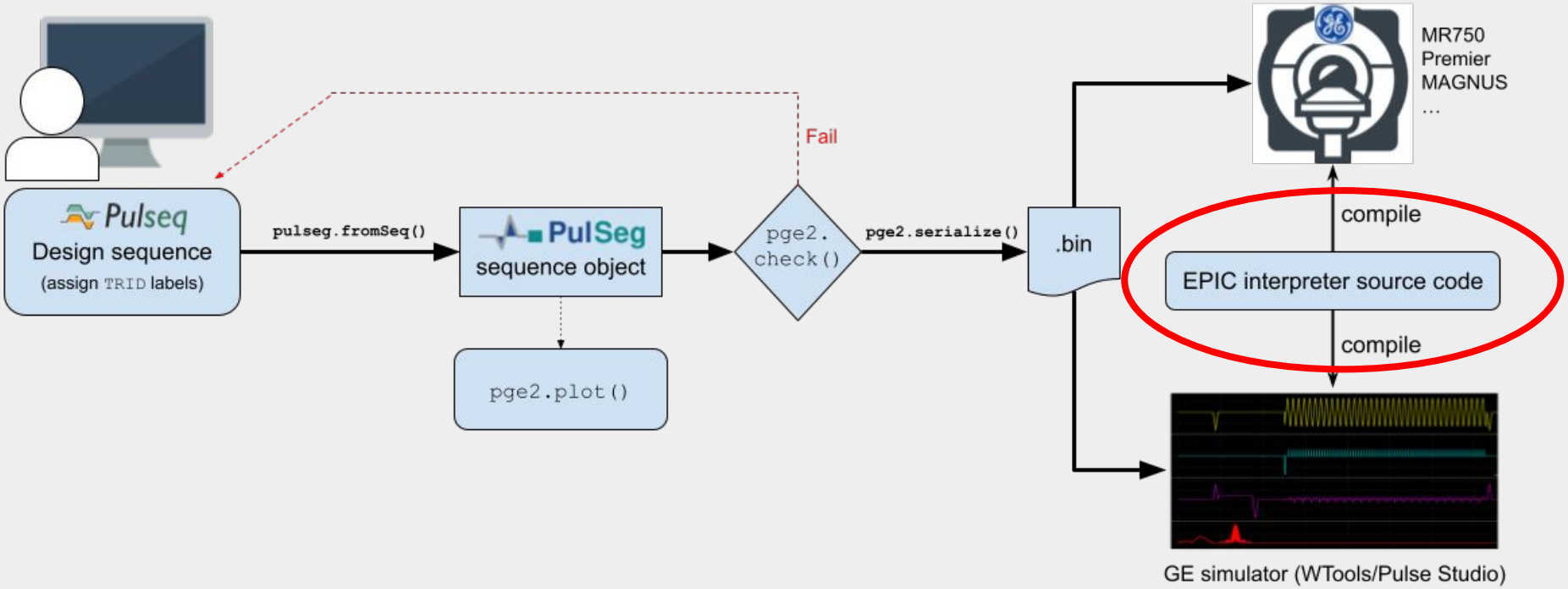
NEW

+pge2

https://github.com/HarmonizedMRI/pge2/

(Formerly part of PulCeq toolbox)

Workflow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Interpreter (pge2)

Download: https://github.com/GEHC-External/pulseq-ge-interpreter

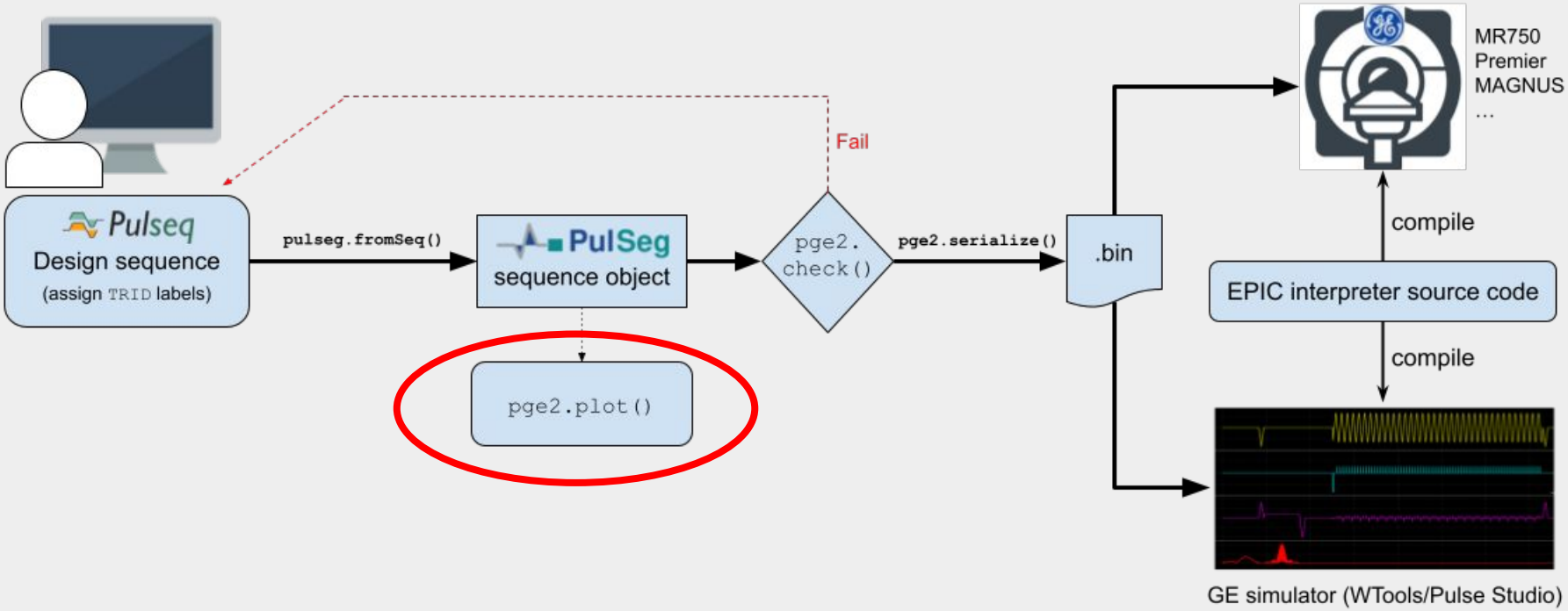
Freely shared within the GE research user community (MIT license)

For access, please contact your local GE representative

For questions, use the **Issues** and **Discussions** forums on Github

Refer to the **Wiki** for usage instructions, troubleshooting, and feature requests

Workflow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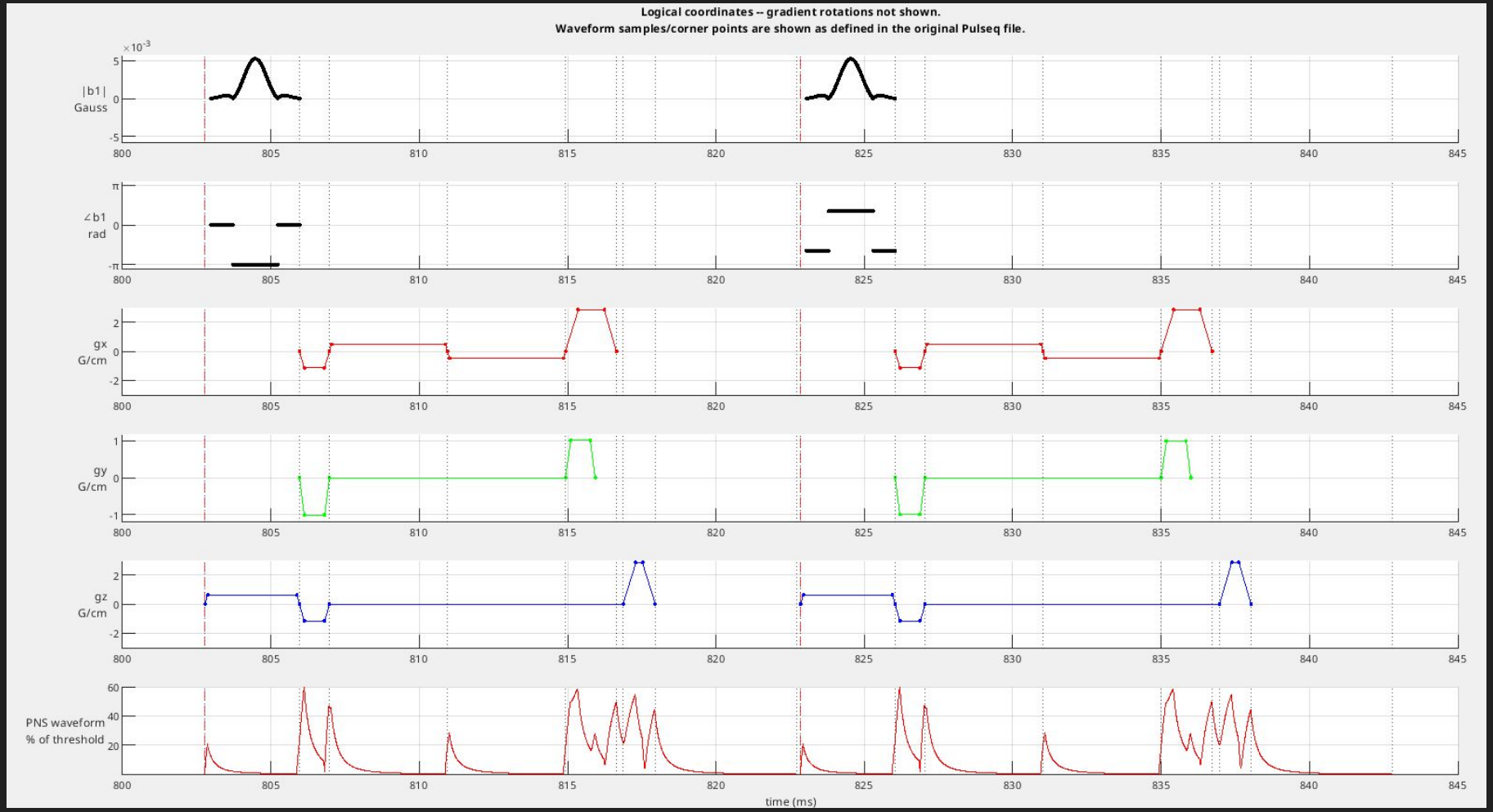
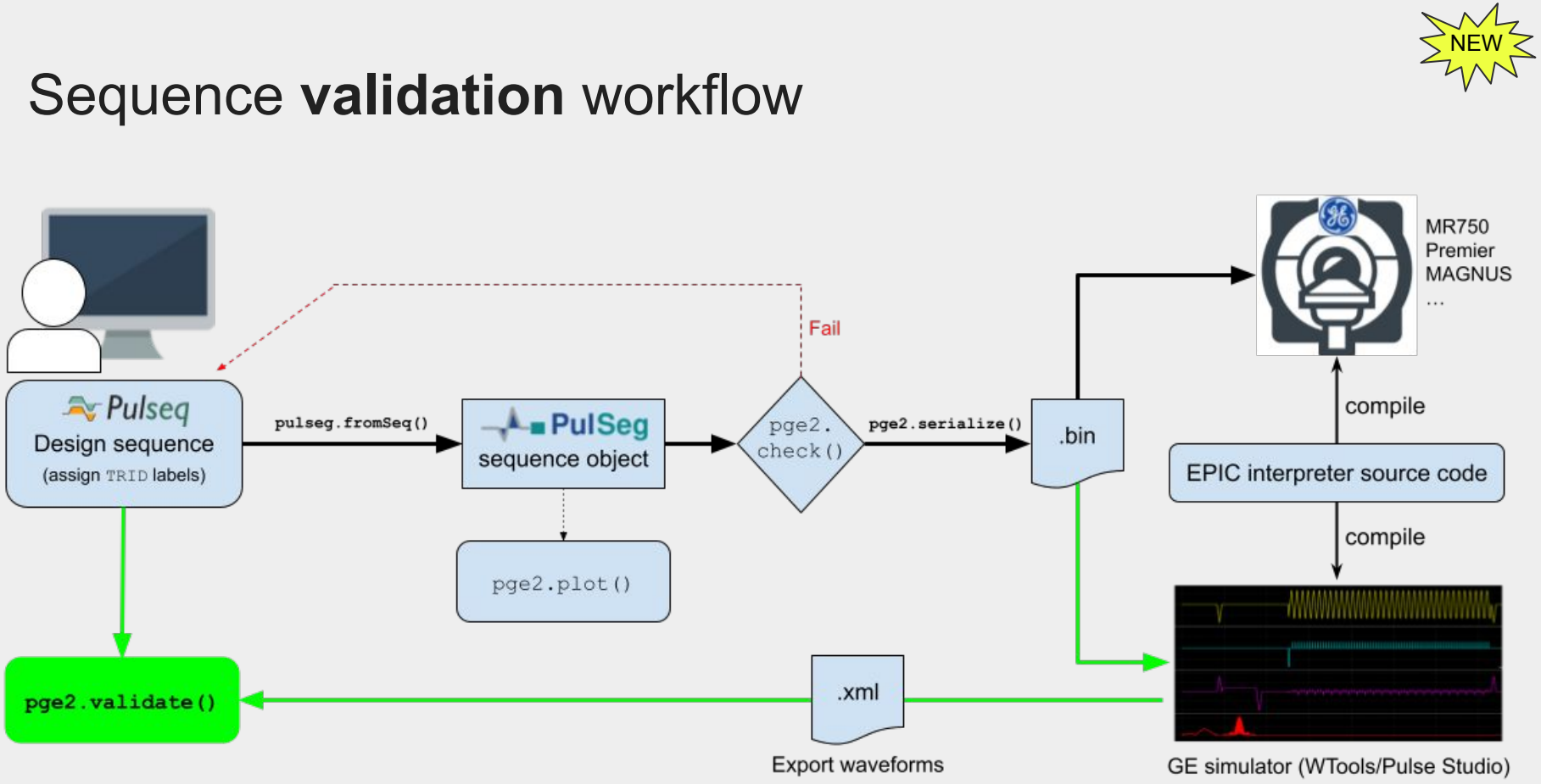
https://github.com/HarmonizedMRI/SequenceExamples-GE/tree/main/pge2

2D GRE



Sequence **validation** workflow



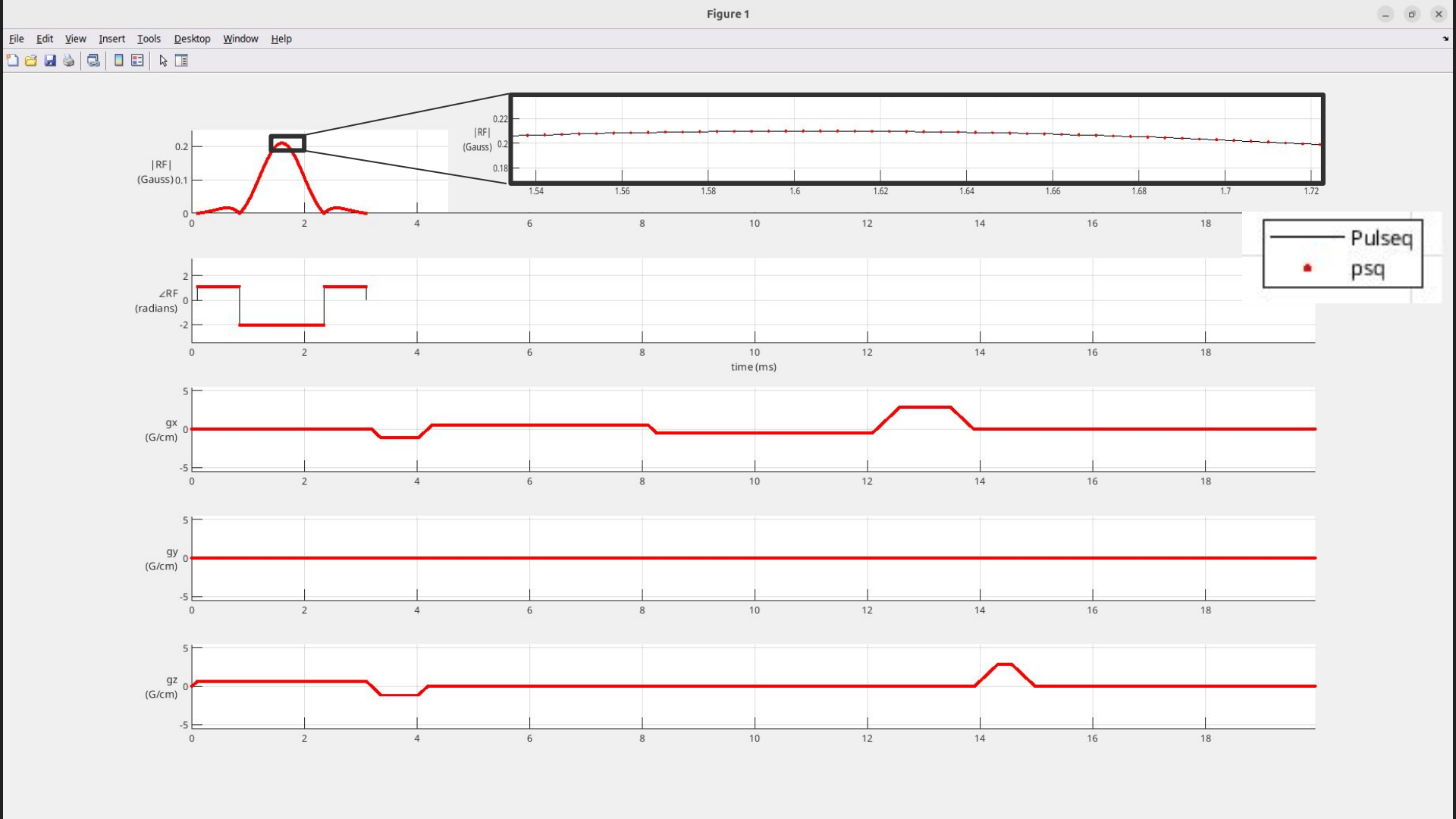


Figure 1
File Edit View Insert Tools Desktop Window Help
|RF| (Gauss)
∠RF (radians)
time (ms)
gx (G/cm)
gy (G/cm)
gz (G/cm)
Pulseq
psq

Low-Level Timing & Segment Rules

Timing rules:

- Rastes times: RF 2 us; Gradient 4 us; ADC 2 us or 1us (MAGNUS)
- Dead/ringdown times can often be ignored (block boundaries ‘disappear’)
- Gradient ramp duration >= 2 samples

Segment rules:

Fixed for all segment instances	Can vary between segment instances
Sequence of base blocks	RF/gradient amplitudes
Block duration (except pure delay blocks)	RF/gradient/ADC phase and frequency offsets
	Duration of pure delay blocks
	Gradient rotation (entire segment -- for now)

Resources

https://github.com/HarmonizedMRI/pulseq

https://github.com/HarmonizedMRI/pge2

Examples: https://github.com/HarmonizedMRI/SequenceExamples-GE/tree/main/pge2

Additional details will be presented in Thursday's talk: "Patient safety and hardware Protection"